

Regime Detection Report

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Project Overview

The goal of this project is to detect market regimes using unsupervised learning on high-frequency order book and trade volume data. The regimes are categorized along three axes:

- **Trending vs. Mean-Reverting**
- **Volatile vs. Stable**
- **Liquid vs. Illiquid**

Using no labeled data, the project applies feature engineering, dimensionality reduction, and clustering algorithms to uncover latent structures in the market.

Custom Features & Engineering

Beyond standard features, several custom, hand-crafted features were introduced to capture nuanced market behavior:

Liquidity & Depth

- **Sloped Depth:** Rate of decay in bid/ask sizes from level 1 to 20. Captures aggressiveness or passivity in order placement.
- **Cumulative Imbalance:** Difference in total bid vs ask volume up to level 20

Volume Features

- **Trade Wipe Level:** Average depth levels cleared by trades in 10s & 30s. Identifies aggressive trade bursts wiping out liquidity.

Volatility & Price Action

- **Rolling VWAP Shift:** Changes in VWAP over 10s, 30s windows
- **Momentum Indicator:** Sum of log returns over 3 consecutive windows

These features were z-score normalized and reduced using PCA (10 components) for effective clustering.

Clustering Methodology

Two clustering techniques were evaluated:

1. KMeans

- Predefined cluster count (k=4)
- Simple and interpretable
- Regimes labeled using median feature values

2. HDBSCAN

- Density-based, does not require preset number of clusters
- Handles noise and non-spherical clusters well

HDBSCAN produced better clustering metrics. KMeans clusters were easier to label and interpret.

Table 1: Clustering Performance Metrics		
Method	Silhouette Score	DB Index
KMeans	0.236	1.495
HDBSCAN	0.387	0.819

Regime Analysis & Labeling

Based on clustering + feature statistics, we defined four primary regimes: Each timestamp in the dataset was

Table 2: Regime Characteristics		
Cluster	Characteristics	Label
0	High spread, high volatility, low liquidity	Volatile & Illiquid
1	Low spread, low volatility, high liquidity	Stable & Liquid
2	Moderate spread, mean-reverting log returns	Mean Reverting
3	Sustained directional movement with liquidity	Trending & Liquid & Stable

tagged with a regime label, enabling time-based and transition analyses.

Visualizations

- **UMAP Projection:** PCA-reduced features were embedded in 2D space using UMAP; clusters showed well-separated structures, validating clustering quality.
- **Regime Evolution:** Time-series plot of regime labels showed transitions and dominance periods.
- **Mid-Price Overlay:** Regime labels overlaid on mid-price series to check correlation with market moves.
- **Regime Transition Matrix:** Showed probabilities of switching from one regime to another.

Conclusion

This project demonstrates how unsupervised techniques can uncover latent regimes in market microstructure data. These regime labels offer valuable insights for downstream tasks like strategy adaptation, risk control, or market surveillance.